

# EVOLUTION OF INVESTMENT INSTRUMENTS: FROM TRADITIONAL FORMS TO RWA TOKENIZATION

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**Abstract.** *The article explores the evolutionary path of investment instruments transformation from their simplest forms to modern digital solutions through a comprehensive historical and systematic analysis spanning from antiquity to the present day. The research identifies and characterizes six key periods in the evolution of investment instruments: natural investment, commercial capital, institutional investment, corporate capital, financial capital, and digital capital. Each period is analyzed through the lens of its characteristic instruments, innovations, limitations, and key transformations, providing a detailed understanding of how investment mechanisms have evolved over time.*

*Particular attention is paid to the analysis of the current stage of investment instruments development, characterized by the implementation of blockchain technologies and RWA (Real World Assets) tokenization. The study reveals the crucial role of technological innovations in democratizing investment processes and expanding access to various asset classes. The authors have systematized the key aspects of investment paradigm transformation according to fundamental criteria such as accessibility, geography, entry threshold, instrument selection, procedures, operation speed, information support, technologies, regulation, and security.*

*The research demonstrates how the convergence of traditional financial systems with blockchain technologies creates a fundamentally new investment paradigm characterized by enhanced accessibility, liquidity, and functionality of investment instruments. The transformative potential of modern investment tools, particularly RWA tokenization, is thoroughly analyzed in the context of the post-war reconstruction of territorial communities' economies. The study emphasizes how these innovations can contribute to the diversification of investment sources for local reconstruction projects, the creation of transparent and targeted funding mechanisms, and the attraction of international investments.*

*The authors present a conceptual vision for the further development of investment mechanisms in the context of digital transformation and post-war recovery needs. The research concludes that the integration of traditional and digital investment instruments, combined with improved regulatory mechanisms and expanded tokenized assets, will enable communities to create technologically progressive economic systems and accelerate the restoration of damaged infrastructure. This approach provides each community with the opportunity to maximize its economic potential by attracting a wide range of investors through innovative digital instruments, collectively contributing to the accelerated recovery and growth of the national economy.*

*The findings of this study have significant implications for both theoretical understanding of investment evolution and practical application in modern economic conditions, particularly in the context of post-war reconstruction and the digital transformation of financial markets.*

**Keywords:** *investing, investment instruments, transformation, RWA-tokenization, digital economy, digital instruments.*

**Problem statement.** In the process of historical development of economic relations, there has been a digital transformation of investment instruments, characterized by the transition from primitive forms of exchange to complex digital financial assets. Analysis of the evolution of investment instruments demonstrates the adaptability of the financial sector to the challenges of the time. Particularly noticeable changes have occurred in recent decades, when the technological revolution launched a new era in the history of investment, introducing the concept of asset tokenization.

Today, we are witnessing a transformation of traditional investment vehicles, opening up unprecedented opportunities for investors of all sizes. The shift from physical assets to digital tokens not only simplifies the investment process, but also democratizes access to financial markets, making them more inclusive and transparent.

**Research objective.** This study aims to systematize and analyze the key stages of the evolution of investment tools, identify the determinants of their transformation, and assess the potential impact of technological innovations on the further development of the investment ecosystem in the context of global digitalization of economic processes.

**Analysis of recent research and publications.** The study of various aspects of the evolution of investment instruments covers a wide range of scientific works, from historical aspects to modern innovations. The historical development of financial markets and investment instruments is deeply explored in the works of Frentrop P., who traces the evolution of corporate governance over four centuries, and Goldsmith R., who analyzes the development of pre-modern financial systems. An important contribution to the understanding of the formation of institutional investors was made by Gelderblom O. and Jonker J. , who studied the Amsterdam securities market of the 17th-18th centuries. The development of the global securities market is covered in detail in the works of Michie R., and the origins of the modern financial revolution are explored in the work of Munro J..

Modern aspects of the development of investment instruments and their transformation in the digital economy are considered in the works of domestic and foreign scientists. In particular, Mints O., Shchepka O., Tolkachova-Mints A. investigate the ethical aspects and prospects for the use of modern and traditional investment instruments. Gavrylina A. analyzes the evolution of scientific thought on the essence of investment as an economic category. A significant contribution to the study of the history of financial capitalism was made by Moshensky S., who examined in detail the processes of its emergence in the pre-industrial era.

The international aspect of the development of investment instruments in the digital economy is highlighted in the works of Dzhusova O.A., Apalkova S.S., who analyze structural shifts in the international capital market, and Casson M., who investigates the development of the theory of internalization in international business. Kovalenko Y., Guley A. and co-authors consider the issue of securitization of financial assets and the reasons for its spread in economic processes. In the context of the development of digital financial instruments, the authors of the article consider the issue of RWA tokenization, exploring this tool as an innovative mechanism for attracting investments and increasing local budget revenues.

However, despite the significant amount of research, an important scientific task is the theoretical and methodological justification of the implementation of innovative investment instruments, in particular RWA tokenization. This highlights the need for in-depth scientific research and systematization of theoretical and practical aspects of the implementation of new forms of investment in the digital economy.

**Research results.** It is advisable to begin the study of the evolution of investment instruments by analyzing the historical prerequisites for their formation. The primary forms of investment, based on barter transactions and the use of precious metals as a means of capital accumulation, laid the foundation for the further development of financial instruments.

When analyzing the complex evolution of investing, it is advisable to consider each historical stage through the prism of its characteristic features and institutional transformations.

The historical transformation of investment instruments took place under the influence of a complex interweaving of economic, social, technological and political factors that shaped the specific conditions of each historical period. Let us consider in more detail the factors of influence at each stage of development. Empirical analysis shows that each stage of development was characterized by specific challenges that stimulated further innovation and improvement.

In the history of the development of investment instruments, fundamental periods can be distinguished, each of which is characterized by specific transformations, challenges, and development factors.

The period of in-kind investment (until the 5th century AD) was characterized by primitive forms of capital investment, where the main instruments were commodity investments in the form of precious metals and land ownership. The defining features of this period were the formation of the first commodity warehouses and the introduction of a system of property registration, but the protest of standardization and high risks of capital loss significantly limited the development of investment activity. Investment remained the prerogative of the elite, and geography was limited to local markets. The key factors of transformation were the basic economic needs of society and the preservation of material values. The development of agriculture created a need for systems of storage and accounting of the harvest, which led to the emergence of the first commodity warehouses. The possibility of protecting property rights led to the formation of the first legal systems of property rights registration.

As early as the 12th century BC, Phoenician and Greek merchants developed a complex system of joint financing of maritime trade. This system involved pooling the capital of several investors to build and equip merchant ships, which allowed for effective risk sharing and increased scale of trade. In ancient Greece, this practice evolved into a specific financial instrument, the bottomry, which became the prototype of modern marine insurance and project finance.

Further development of investment mechanisms was observed during the Roman Empire, where the first public companies – *societas publicanorum* – emerged. These organizations demonstrated characteristics that bring them closer to modern joint-stock

companies: the possibility of transferring ownership shares, limited liability of investors, professional management and the public nature of their activities. *Societas publicanorum* carried out a wide range of government tasks, including collecting taxes, building public buildings, developing mines and providing the army with the necessary resources. It is important to note that these companies represented a significant evolutionary step from simple partnerships to more complex forms of collective investment, in fact laying the foundations of modern public-private partnerships.

The next period of development of investment instruments (V-XV centuries) can be called the period of commercial capital, which was marked by significant progress in the development of financial instruments due to the emergence of bank deposits and the formation of marine insurance. The creation of banks and international trade networks contributed to the expansion of the geography of investment, but religious restrictions and the issuance of uniform standards created significant obstacles. The key transformation was the emergence of financial intermediaries and the standardization of document flow, which laid the foundations for the development of commercial law. The development of cities and trade routes created a demand for banking services, which led to the formation of the banking system. The formation of trade guilds responded to the need to organize and protect the interests of merchants.

The Middle Ages were characterized by significant transformations in the field of financial and investment instruments, which laid the foundation for the further development of the modern financial system. A particularly important role in this process was played by the Italian city-states, which became centers of innovation in the financial sphere. The most significant contribution of this period was the development of the *commendam* system - a special form of commercial partnership that arose in the Mediterranean trading cities, primarily in Venice and Genoa.

A *commendam* was a medieval form of commercial partnership, common in the Mediterranean countries, particularly in the Italian city-states. It involved a contractual relationship between an active partner (merchant-sailor), who organized and conducted trade operations, and a passive partner (investor), who provided capital to finance trading expeditions without taking part in the activities themselves. Profits and risks

were shared in predetermined proportions: in the event of a successful trade, the profits were shared between the partners, and in the event of the loss of the ship or goods, the active partner was usually not financially liable, except in cases of fraud or negligence. This model became the basis for the development of modern venture capital investments and limited partnerships.

Parallel to the development of the command, banking emerged. Italian banking houses, among which the Medici held a special place, developed a double-entry bookkeeping system and introduced the use of bills of exchange (*lettere di cambio*). These financial instruments allowed international trade transactions to be carried out without the physical movement of gold or silver, which significantly reduced risks and transportation costs.

This period also saw the development of the first forms of maritime trade insurance. Marine insurance emerged as a separate contract, separate from the loan, which was an important step in the development of financial instruments. Specialized insurance contracts emerged to protect against various types of risks associated with maritime trade.

Medieval religious orders, especially the Knights Templar, made significant contributions to the development of the financial system, creating a complex international network of financial services. They provided safekeeping services, made money transfers between different regions, and issued documents similar to modern letters of credit.

In the late medieval period, the first prototypes of joint-stock companies began to take shape, especially in the mining industry of Central Europe. Mining enterprises began to issue *kuxe*, which were shares in the ownership of mining enterprises, which began to be issued in Saxony and Bohemia in the Middle Ages. They were analogues of modern shares and allowed to raise capital for the extraction of minerals. The owners of *kuxe* became co-owners of the mine and were entitled to a share of the profits derived from the extraction of ore. These shares could be bought, sold or inherited, which made them one of the early examples of marketable financial instruments. The

kuxe system played an important role in the development of equity financing and capital investment in industry.

The medieval period was thus a time of important innovations in the field of financial instruments. The foundations of many modern financial concepts and institutions were laid, including banking, insurance, corporate law, and international financial transactions. These innovations laid the foundation for the further development of the financial system in the Renaissance and Modern times.

The period of institutional investment (15th-18th centuries) brought fundamental changes through the emergence of company shares and government bonds. The creation of stock exchanges and the introduction of a quotation system, although they significantly expanded investment opportunities, high entry costs and information asymmetry remained significant limitations. The need to finance large-scale maritime expeditions led to the emergence of joint-stock companies as a mechanism for raising capital. The growth of international trade stimulated the development of stock exchanges and the creation of a quotation system. The formation of colonial empires required new financial instruments to manage significant trade flows.

A key innovation of this period was the creation of the first modern-style joint-stock companies. The British East India Company, founded in 1600, and the Dutch East India Company (VOC), founded in 1602, pioneered a new form of business organization. Most importantly, these companies introduced permanent capital and transferable shares, which fundamentally distinguished them from medieval trading companies. This innovation allowed investors to pool their capital and spread the risks across entire trading expeditions to India, rather than having individual wealthy individuals finance individual projects. This mechanism helped to shape modern capital markets and democratize asset ownership, as participation in the company became available to a wider range of investors, not just the elite. The increase in the number of shareholders contributed to a significant increase in the value of the VOC, which exceeded the combined value of the 20 largest modern companies. The Dutch East India Company was the first to introduce regular dividend payments to shareholders, which created a new standard for investment activities.

A significant achievement of this period was the establishment of the Amsterdam Stock Exchange in 1602, the world's first continuously operating stock exchange for trading securities. The exchange provided a liquid market for trading stocks and bonds, which contributed to the development of the secondary securities market. The exchange also introduced the first forms of futures contracts and options, which laid the foundation for the development of the derivatives market.

During this period, the banking system developed significantly. The first state banks emerged, such as the Bank of Amsterdam (1609), which became a model for the later development of central banks. The bank introduced a system of non-cash payments and pioneered the standardization of monetary circulation. Private banking houses also developed, especially in Italian cities, which expanded the range of financial services and improved risk management methods.

An important innovation was the emergence of new forms of financing maritime trade. The system of bottomry bonds developed - a special type of maritime loan, where the ship served as collateral. More complex forms of marine insurance also appeared, allowing for a more efficient distribution of the risks of maritime trade. The first modern insurance companies, specializing in marine insurance, emerged.

In the field of public finance, there were significant innovations related to the issuance of government bonds. Particularly important were the experiences of the Republic of Venice and the Dutch Republic, which created effective systems of public debt. These innovations allowed states to raise significant funds to finance military expeditions and colonial expansion.

The period was also characterized by the development of complex financial instruments. The first futures contracts appeared on commodity exchanges, especially in the tulip trade in Holland.

An important aspect of this period was the formation of the legal framework for the functioning of joint-stock companies and financial markets. The concepts of limited liability, corporate governance, and protection of shareholder rights developed. These legal innovations created the basis for the further development of corporate law and financial regulation.

Thus, the Renaissance and the Great Geographical Discoveries (15th-18th centuries) were a period of fundamental innovations in the financial sphere. The main institutions of the modern financial market were created: joint-stock companies, stock exchanges, complex financial instruments. These innovations laid the foundation for the further development of the global financial system and largely determined the nature of modern financial markets.

The next stage of the transformation of investment instruments took place in the 18th and 19th centuries. The Industrial Revolution was a period of fundamental transformations in the development of investment instruments and the financial system as a whole. The rapid development of industry and transport created an unprecedented demand for capital, which stimulated the emergence of new forms of financing and investment.

The development of the railway industry, which required large-scale investments, was particularly significant. This led to the widespread use of joint-stock companies and improved mechanisms for public offering of shares. Railway companies became the first large corporations of the modern type that actively attracted funds from small investors. This process was accompanied by the development of the secondary securities market and the improvement of stock exchange trading.

The bond market, especially corporate bonds, developed significantly. Railway and industrial companies actively issued bonds to finance long-term projects. This contributed to the standardization of debt instruments and the development of methods for assessing credit risk. The first rating agencies appeared to assess the reliability of bond issues.

During this period, investment banking emerged as a separate area of financial activity. Leading banking institutions such as Rothschild, JP Morgan, and Goldman Sachs made a significant contribution to the development of investment banking. Their specialization included the issuance and placement of securities, as well as the structuring of mergers and acquisitions. These financial institutions introduced innovative underwriting methodologies and developed syndicated lending mechanisms, which ensured effective risk sharing in the implementation of large-scale

financial transactions. The implementation of these financial instruments contributed to the formation of the modern architecture of investment banking and laid the methodological foundations for managing financial risks in large transactions on the capital market.

An important innovation was the emergence of the first investment funds (investment trusts) in the UK. These structures allowed small investors to diversify their investments through professionally managed portfolios of securities. The concept of collective investment quickly spread and became an important element of the financial system.

The period was also marked by the development of futures markets, especially in agricultural commodities. The Chicago Mercantile Exchange, founded in 1848, pioneered the standardization of futures contracts and the development of clearing mechanisms. This greatly improved the efficiency of risk management in agriculture and trade.

During this period, the formation of the modern banking system took place. Central banks emerged in many countries, which took on the functions of regulating monetary circulation and banking supervision. A system of commercial banks developed, which provided a wide range of financial services for industry and trade.

An important aspect was the improvement of corporate law and financial regulation. Laws were passed to regulate the activities of joint-stock companies, protect the rights of investors and establish standards for financial reporting. The Joint-Stock Companies Act of 1856 introduced the mandatory maintenance of a register of shareholders and the issuance of share certificates, which created a legal basis for documentary confirmation of ownership rights. Further improvement of the regulatory framework occurred with the adoption of the Companies Act of 1862, which established a clear procedure for the transfer of ownership of shares through a system of registration in the company's books.

Thus, the Industrial Revolution was a period of formation of the main elements of the modern financial system. Key investment instruments were created and improved, financial institutions and market regulation mechanisms developed.

Innovations of this period laid the foundation for the further development of global financial markets in the 20th century.

The 20th century was marked by the diversification of investment instruments through the emergence of mutual funds, ETFs, and derivatives. The introduction of electronic trading and global markets was accompanied by technological limitations and regulatory barriers. The development of economic science led to the formation of the theory of portfolio investment and the principles of diversification. The advent of computers revolutionized trading processes and transactional accounting. The emergence of the social security system created a demand for pension funds and long-term investment products. This period can be called the period of financial capital.

The transformation of investment instruments in the 20th century was characterized by fundamental changes in the financial system and a significant expansion of investment opportunities. The formation of the modern financial market was influenced by several key factors. First of all, scientific and technological progress and the introduction of computer technologies radically changed the mechanisms of trading, providing the possibility of conducting fast transactions and analyzing large data sets. This created the prerequisites for the development of algorithmic trading and the emergence of new types of financial instruments.

One of the key achievements was the emergence of modern stock markets. The New York Stock Exchange became the world's leading financial center. Stock exchange operations were standardized, uniform rules for trading and settlements were introduced. An important innovation was the system of margin trading, which allowed investors to purchase securities with partial leverage.

In 1924, the first modern mutual fund, the Massachusetts Investors Trust, was created in the United States. This launched the mutual fund industry, which made stock market investments accessible to a wide range of small investors. Mutual funds offered professional investment management and diversification opportunities even for small investment amounts.

The stock market crash of 1929 and the subsequent Great Depression led to fundamental changes in the system of financial regulation. In the United States, the

Securities Act of 1933 and the Securities Exchange Act of 1934 were passed, and the Securities and Exchange Commission (SEC) was created. These innovations established standards for disclosure by issuers, rules for securities trading, and mechanisms for protecting investor rights.

A significant achievement was the emergence and development of pension funds as institutional investors. This created a new channel for long-term investment in the economy and contributed to the development of the corporate bond market. Pension funds have become important participants in financial markets, forming significant investment portfolios.

An important aspect of this period was the development of investment analysis and financial theory. The first systematic approaches to securities valuation appeared, in particular Benjamin Graham's work on fundamental analysis. The foundations of modern investment theory and risk management were laid.

Securities accounting systems in Europe underwent changes with the introduction in the UK of a legislative framework for dematerialization through the Uncertificated Securities Regulations (USR) in 1995. This regulation became a fundamental factor that legitimized electronic records as legally significant evidence of ownership of securities.

A key outcome of this regulatory transformation was the creation of the Certificateless Registry for Electronic Share Transfer (CREST) in 1996, the first central securities depository in the UK and Ireland. CREST's institutional role was defined by several fundamental aspects of its functioning:

1. The introduction of the Delivery versus Payment (DvP) mechanism has significantly reduced the systemic risks inherent in previous settlement methods without synchronizing the transfer of assets and funds (Free of Payment (FoP)).
2. CREST's regulatory status as an officially recognized and supervised intermediary ensured legal certainty and irreversibility of transactions in the securities market.

3. The depository's technological infrastructure allowed for optimized communication between market participants, ensuring a two-hour ownership confirmation cycle through interaction with brokers and registrars.

4. Integration with the central bank's settlement system in 2001, which strengthened the role of the depository as a key element of the country's financial infrastructure.

It is also worth noting that securitization, an innovative financial mechanism that allows illiquid assets to be converted into liquid securities, played a special role in transforming the investment landscape. This process began in the 1970s with the securitization of mortgage loans in the United States and later spread to other types of assets, including auto loans, credit cards, and lease payments. Securitization significantly expanded the possibilities for investment and risk redistribution, although its excessive use later became one of the factors of the 2008 financial crisis.

The development of portfolio investment theory, pioneered by Harry Markowitz, led to fundamental changes in approaches to investment management. The concept of portfolio diversification and optimization became the theoretical basis for the creation of mutual funds and ETFs (Exchange-Traded Funds), which gave individual investors access to professionally managed diversified portfolios. In parallel, the derivatives market developed, offering tools for hedging risks and speculative trading.

During this period, the derivatives market developed rapidly. The creation of the Chicago Board Options Exchange (CBOE) in 1973 and the development of the Black-Scholes formula for option pricing were key moments in the development of derivatives. New risk management tools emerged: financial futures, swaps, structured products.

An important innovation was the emergence of venture capital as a separate form of investment. The first venture capital funds, created in the 1960s, provided financing for technological startups and innovative companies. This created a new mechanism for financing entrepreneurship and contributed to the development of high-tech industries.

The 1980s saw the emergence of the private equity industry, which specialized in leveraged buyouts and business restructurings, creating a new segment of the investment market and new opportunities for institutional investors.

Thus, the middle and end of the 20th century became a period of fundamental innovations in the field of investment instruments and investment theory. New asset classes were created, risk management mechanisms developed, and new approaches to portfolio investment were formed. These innovations laid the foundation of the modern global financial system and determined the directions of its further development in the 21st century.

The technological revolution in the financial sector has led to the democratization of investing through the emergence of online brokers and trading platforms. This has significantly reduced transaction costs and barriers to entering financial markets for private investors. At the same time, the increasing complexity of financial instruments and market mechanisms has required improved risk management systems and strengthened regulatory oversight to ensure the stability of the financial system.

The 21st century is characterized by a radical transformation of investment instruments under the influence of the digital revolution and global economic changes. The key trends have been the digitalization of financial markets, the emergence of fundamentally new asset classes, and the formation of innovative investment strategies. The defining trend is the development of algorithmic and high-frequency trading. Computer algorithms perform millions of operations in microseconds, which has radically changed the pricing mechanisms in financial markets. Artificial intelligence and machine learning have allowed the creation of complex models for forecasting and managing investment portfolios. This period can be called the period of digital capital.

Distributed Ledger Technology (DLT) has become a fundamental innovation that has changed the paradigm of financial transactions and data storage. DLT has enabled the creation of decentralized systems where transaction information is stored simultaneously on many network nodes, significantly increasing the security and transparency of operations.

The introduction of the Ethereum platform (2015) has brought about significant changes in the development of blockchain technologies, expanding the functionality of digital assets from simple transactions to programmable financial instruments. A key innovation has been the implementation of smart contracts - automated algorithmic protocols that select certain operations under specified conditions. The main factors transforming the investment environment have been: automation of financial transactions through smart contracts that ensure transaction execution, data verification and asset management without the participation of intermediaries; tokenization of assets, which has enabled the digital representation of a wide range of tangible and intangible assets, including paper prices, real estate, and intellectual property; democratization of investment through the introduction of a fractional ownership mechanism, which has broken down barriers to entry into previously inaccessible investment markets. The result of these innovations has been the formation of a decentralized financial ecosystem, where traditional trust mechanisms have been replaced by cryptographic algorithms and program code.

Of particular importance in the transformation of the investment landscape is the tokenization of real world assets (RWA), which allows the representation of ownership rights to physical assets in the form of digital tokens on the blockchain. This process democratizes access to traditionally illiquid assets, such as real estate, art, or rare metals, allowing retail investors to participate in markets previously inaccessible to them. RWA tokenization also significantly increases the liquidity of these assets and reduces transaction costs.

The development of artificial intelligence and Big Data technologies has led to the emergence of algorithmic trading and automated investment advisors (robo-advisors). These technologies allow analyzing huge amounts of data in real time, identifying hidden patterns, and creating personalized investment strategies. The introduction of blockchain-based smart contracts has automated the execution of contract terms and reduced counterparty risks.

The emergence of decentralized finance (DeFi) has created an alternative financial ecosystem that operates without traditional intermediaries. DeFi protocols

offer a wide range of financial services, including lending, trading, and insurance, implemented through smart contracts. At the same time, the development of micro-investment platforms and shares that are divided into parts has made investing accessible to a wider range of participants.

The integration of mobile technology and financial services has led to the emergence of neobanks and investment applications that provide 24/7 access to global markets. Biometric authentication and multi-factor authentication technologies have increased the security of digital transactions. The development of quantum computing creates both new opportunities for optimizing investment strategies and challenges for existing cryptographic protection systems.

Regulators are adapting to new technological realities by developing regulatory frameworks for digital assets and introducing the concept of regulatory sandboxes for testing innovative financial products. The growth of cyber threats is stimulating the development of advanced security systems and the formation of new standards for protecting investment platforms. The use of Zero-Knowledge Proofs technology allows ensuring the confidentiality of transactions while maintaining their verifiability.

Thus, the evolution of investment instruments demonstrates a complex interaction of technological, economic and social factors, with each new stage characterized by the emergence of more complex and diversified forms of investment, which compensates for the increasing needs of society and the challenges of the time.

In the context of the historical analysis of the evolution of investment instruments, it is advisable to systematize the key characteristics of each period to form a holistic understanding of paradigmatic shifts in the field of investment. Table 1 presents a comprehensive analysis of the transformation of investment processes, showing the specifics of each stage through the prism of the tools used, existing limitations and defining changes that laid the foundation for the further development of investment activities. The proposed systematization allows us to trace the evolutionary path from traditional forms of investment to modern digital instruments and outline the main trends in the transformation of investment processes.

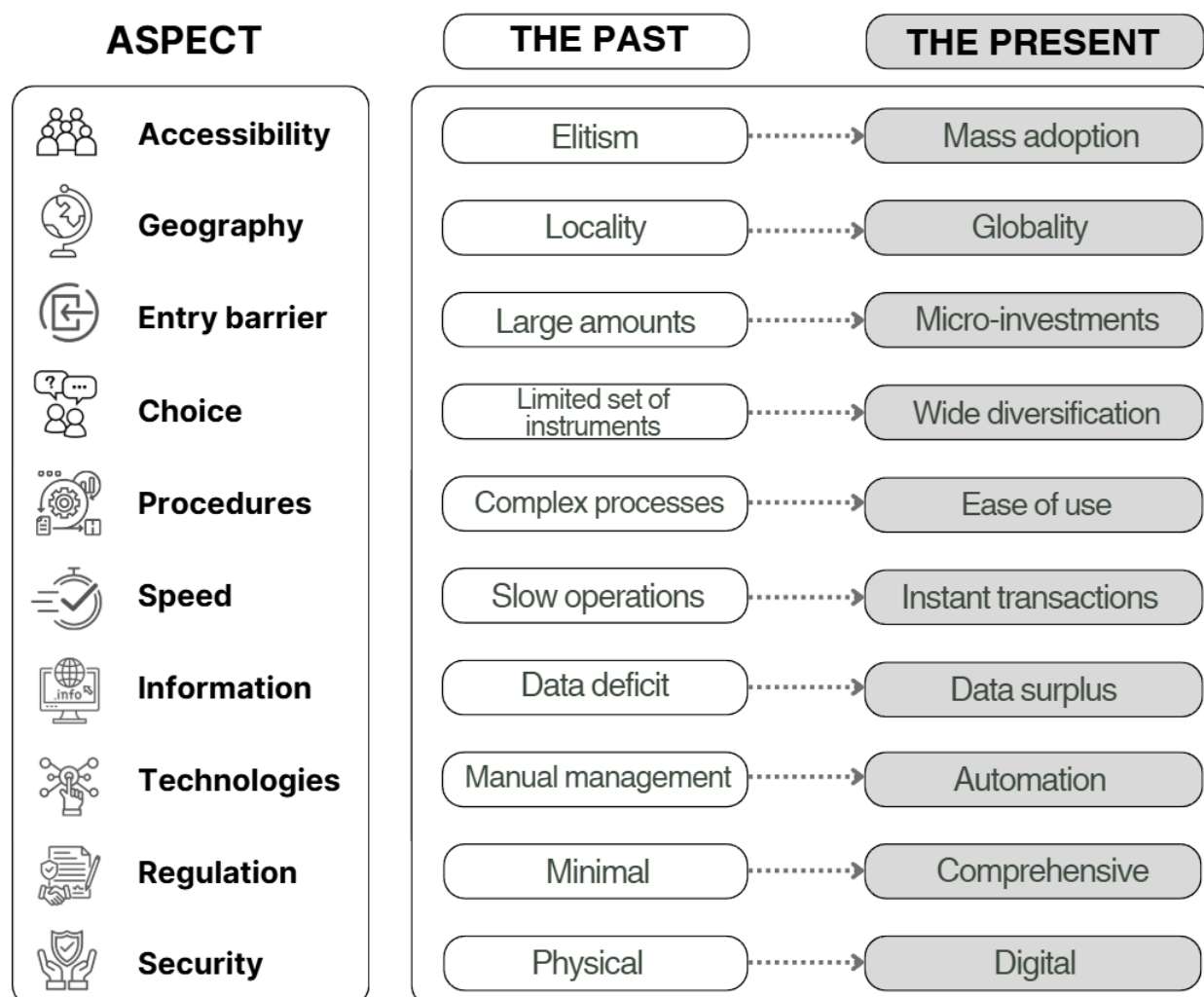
Table 1.

### Paradigmatic transformation of investment processes: historical and systemic analysis\*

| Period                                                      | Tools and forms                                                                                                                                                        | Features and innovations                                                                                                                                                                                                                          | Limitations and challenges                                                                                                                                                                                              | Key transformations                                                                                                                                                                                                                             |
|-------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Period of in-kind investment<br>(before the 5th century AD) | <ul style="list-style-type: none"> <li>Commodity investments (gold, silver)</li> <li>Secured loans</li> <li>Land holdings</li> <li>Trading operations</li> </ul>       | <ul style="list-style-type: none"> <li>The first warehouses</li> <li>The emergence of debt instruments</li> <li>Rights registration system</li> <li>Basic forms of insurance</li> <li>Marine loans</li> <li>The first public companies</li> </ul> | <ul style="list-style-type: none"> <li>Lack of standardization</li> <li>High risks of loss</li> <li>The need for large capital</li> <li>Lack of protection of rights</li> </ul>                                         | <ul style="list-style-type: none"> <li>Availability: elite only</li> <li>Geography: local</li> <li>Adjustment: minimal</li> <li>Security: physical security</li> </ul>                                                                          |
| Commercial capital period<br>(5th-15th centuries)           | <ul style="list-style-type: none"> <li>Bank deposits</li> <li>Trading companies</li> <li>Marine insurance</li> <li>Bill of exchange transactions</li> </ul>            | <ul style="list-style-type: none"> <li>Creation of banks</li> <li>International retail chains</li> <li>Insurance development</li> <li>Bill of exchange system</li> </ul>                                                                          | <ul style="list-style-type: none"> <li>The need for connections</li> <li>Religious restrictions</li> <li>Lack of uniform standards</li> <li>Political risks</li> </ul>                                                  | <ul style="list-style-type: none"> <li>The emergence of financial intermediaries</li> <li>Geographical expansion</li> <li>Document standardization</li> <li>Development of commercial law</li> </ul>                                            |
| Institutional investment period<br>(15th-17th centuries)    | <ul style="list-style-type: none"> <li>Company shares</li> <li>Government bonds</li> <li>Stock market instruments</li> <li>Banking products</li> </ul>                 | <ul style="list-style-type: none"> <li>Creation of exchanges</li> <li>Quotation system</li> <li>Dividend payments</li> <li>Brokerage services</li> <li>Futures contracts</li> </ul>                                                               | <ul style="list-style-type: none"> <li>High entrance fee</li> <li>Limited access</li> <li>Information asymmetry</li> <li>Monopoly</li> </ul>                                                                            | <ul style="list-style-type: none"> <li>Formation of capital markets</li> <li>The emergence of public companies</li> <li>Development of mediation</li> <li>Standardization of securities</li> </ul>                                              |
| Corporate Capital Period<br>(18th-19th centuries)           | <ul style="list-style-type: none"> <li>Shares of industrial companies</li> <li>Corporate bonds</li> <li>Investment funds</li> <li>Banking products</li> </ul>          | <ul style="list-style-type: none"> <li>Clearing system</li> <li>Professional management</li> <li>Investment diversification</li> <li>International investments</li> </ul>                                                                         | <ul style="list-style-type: none"> <li>Difficulty of access</li> <li>High commissions</li> <li>Slow calculations</li> <li>Limited liquidity</li> </ul>                                                                  | <ul style="list-style-type: none"> <li>Industrialization of finance</li> <li>Development of stock markets</li> <li>The emergence of collective investments</li> <li>International integration</li> </ul>                                        |
| Financial capital period<br>(20th century)                  | <ul style="list-style-type: none"> <li>Mutual funds</li> <li>ETFs</li> <li>Derivatives</li> <li>Pension products</li> <li>Securitized assets</li> </ul>                | <ul style="list-style-type: none"> <li>Electronic bidding</li> <li>Global markets</li> <li>Index investing</li> <li>Automation of operations</li> <li>Development of securitization</li> </ul>                                                    | <ul style="list-style-type: none"> <li>Technological and regulatory barriers</li> <li>Information asymmetry</li> <li>Complexity of new tools</li> </ul>                                                                 | <ul style="list-style-type: none"> <li>Computerization of trade</li> <li>Globalization of markets</li> <li>Development of regulation</li> <li>Institutionalization of investment</li> <li>Asset securitization</li> </ul>                       |
| The era of digital capital<br>(21st century)                | <ul style="list-style-type: none"> <li>Online investments</li> <li>Cryptocurrencies</li> <li>Tokenized assets</li> <li>P2P platforms</li> <li>Robo-advisors</li> </ul> | <ul style="list-style-type: none"> <li>Instant transactions</li> <li>Microinvesting</li> <li>Smart contracts</li> <li>Personalization of strategies</li> <li>RWA tokenization</li> <li>Regulatory sandboxes</li> </ul>                            | <ul style="list-style-type: none"> <li>Cybersecurity</li> <li>Information overload</li> <li>Volatility</li> <li>Technological risks</li> <li>Regulatory uncertainty</li> <li>Financial and computer literacy</li> </ul> | <ul style="list-style-type: none"> <li>Democratization of access</li> <li>Digitalization of processes</li> <li>Global availability</li> <li>Personalization of services</li> <li>Blockchain technologies</li> <li>Asset tokenization</li> </ul> |

\*Source: developed by the authors

If we summarize the above-mentioned features of the transformation of the investment paradigm, we can illustrate how, over time, key aspects of the investment landscape have changed due to the development of technology, globalization, and democratization of financial markets (Fig. 1).



**Fig. 1. Key aspects of investment transformation\***

*\*Source: developed by the authors*

In the past, investing was only accessible to a limited circle of individuals, characterized by locality and elitism. Large amounts of capital were required to participate in the market, which limited access for the general population. The range of investment instruments was limited, and investment processes were complex and slow. Information about markets and instruments was difficult to access, which made it difficult to make informed decisions. Investment management was manual, with minimal use of technology. Regulation and security were fragmented, which increased risks for investors.

The current stage of development of investment instruments is determined by the mass adoption of digital technologies and global integration of financial markets. RWA tokenization and digital platforms play a key role in expanding micro-investing opportunities, ensuring the accessibility of investments to a wide range of participants, regardless of the size of their capital.

Tokenization of real assets helps to increase liquidity and creates conditions for broad diversification of portfolios, which minimizes investment risks. Digital platforms simplify the investment process thanks to automated transactions, intuitive interface and operational processing of operations. The abundance of information resources provides investors with access to detailed analytical data, which helps to make informed decisions.

Automation of investment management allows you to optimize time and financial costs, increasing the efficiency of resource allocation. The introduction of comprehensive digital regulation and modern cybersecurity mechanisms guarantees a high level of protection of assets and investors' interests, which is a key factor in the development of the global investment environment.

RWA tokenization has not yet been implemented in Ukraine, but it is a very promising tool that has the potential to effectively transform the mechanisms for financing community development projects and attracting investment.

Factors hindering the implementation of RWA tokenization in Ukraine include the lack of proper regulatory framework for asset tokenization, technical and infrastructure limitations associated with the implementation of blockchain solutions at the local level, and lack of community awareness of the benefits and capabilities of this technology.

At the same time, the successful implementation of RWA tokenization has the potential to significantly impact the further development of the investment ecosystem in Ukraine. The use of this technology can contribute to attracting investments in the restoration and development of communities, ensuring transparency and efficiency of asset management. In addition, RWA tokenization can stimulate the development of innovative financial instruments and platforms that will allow attracting funds from a wide range of investors, including international ones. Thus, the implementation of

RWA tokenization can be an important step towards the formation of a modern and effective investment ecosystem in Ukraine, which will contribute to the sustainable development of communities and economic growth of the country as a whole.

To realize this potential, joint efforts of the state, business, and civil society are needed to create a favorable regulatory environment, raise awareness of the benefits of RWA tokenization, and build the necessary technical infrastructure.

**Conclusion.** The study of the evolution of investment instruments demonstrates a fundamental transformation of investment mechanisms and forms – from traditional tangible assets to tokenized digital assets representing real assets. The traced evolutionary trajectory demonstrates the gradual overcoming of historical limitations of investment activity through the introduction of technological innovations that have significantly expanded the accessibility, liquidity and functionality of investment instruments.

The key trend of the current stage of development is the convergence of traditional financial systems with blockchain technologies, which creates a fundamentally new investment paradigm. The introduction of RWA tokenization can become an effective tool for modernizing investment processes in communities. Through the use of blockchain technologies and smart contracts, RWA tokenization provides transparency, liquidity and automation of asset management. This allows you to attract a wider range of investors, including international ones, and diversify sources of financing for local reconstruction projects.

The practical application of RWA tokenization in communities can contribute to the creation of an innovative economic system that is more resilient and adaptive to the challenges of the post-conflict period. Thanks to digital tools, communities will be able to more effectively raise funds for infrastructure restoration, create new jobs, and stimulate economic development at the local level.

In addition, the use of RWA tokenization will ensure transparent and targeted use of investment funds. Thanks to blockchain technology, all transactions will be recorded in an immutable register, which minimizes the risks of corruption and misuse of funds. This will increase investor confidence and help attract more investment in community development projects.

Thus, the implementation of RWA tokenization has significant practical potential for the recovery and development of the Ukrainian economy in the post-war period. Thanks to innovative investment tools, communities will be able to more effectively attract funds, accelerate economic growth and create new opportunities for local development. This will contribute to strengthening the economic sustainability and competitiveness of Ukraine in the international arena.

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